



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2008
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 100

- Attempt all questions.
- All questions are of equal value.

Examiner: *F.Nesbitt*

SECTION A

QUESTION 1(16 marks)

- (a) Find $\frac{d}{dx}\cos 3x$ 1
- (b) Convert 36° to radians. Give your answer in terms of π . 1
- (c) For the series
 $2 + 9 + 16 \dots\dots$ find: 3
- (i) the 12th term
- (ii) the sum of the first 15 terms
- (d) State the Domain and Range of $y = \sqrt{5-2x}$ 2
- (e) A sector of a circle with radius 5 cm subtends an angle of 50° at the centre.
Find the area of the sector in terms of π . 2
- (f) Differentiate the following with respect to x :
- (i) $\frac{1}{2}\sin x - \cos 2x$ 1
- (ii) $\sin^2 2x$ 2
- (iii) $e^{3x-1} - \frac{1}{2}e^{-2x-5}$ 2
- (iv) $\frac{2x}{\cos 2x}$ 2

QUESTION 2 (17 marks)

- (a) Evaluate: $\sum_{n=1}^{20} (3n - 4)$ 2
- (b) Express $0.\dot{4}3$ as a simplified fraction. 2
- (c) Find the value of x if $3x$, $2x + 4$ and $6x - 2$ are consecutive terms of an arithmetic series. 2
- (d) Find a primitive function of $\cos 3x - \sin 2x$ 1
- (e) If $f'(x) = 4(3x - 4)^3$ and $f(2) = 3$, Find $f(x)$ 2
- (f) For the function $y = x^3 - 3x - 4$
- (i) Find the co-ordinates of any stationary points and determine their nature. 4
 - (ii) Find any point(s) of inflexion 1
 - (iii) Sketch the curve showing all important features 2
 - (iv) For which values of x is the curve concave up? 1

SECTION B (start a new booklet)

QUESTION 3 (18 marks)

Find

(a)

(i) $\int (2x - 3)(x - 1) dx$ 2

(ii) $\int (4x - 1)^3 dx$ 2

(iii) $\int_0^1 \frac{x^3 + 2x^2 - x}{x} dx$ 2

(iv) $\int_1^2 \frac{1+x}{\sqrt{x}} dx$ 2

(b) In a certain series $S_n = n^2 - 2n$

(i) Find Term 1 1

(ii) Find the sum of the first two terms 1

(iii) Show that this is an arithmetic series. 2

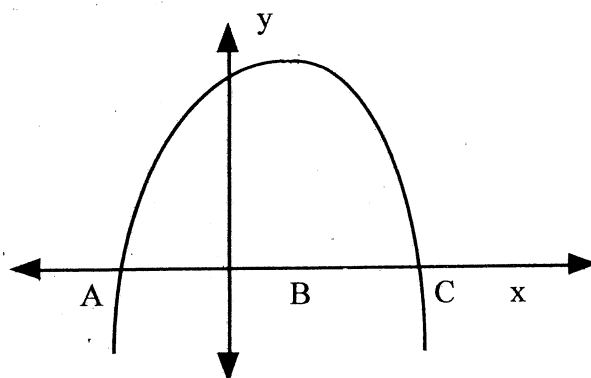
(iv) Write an expression for T_n . 1

(c) (i) On the same set of axes, sketch the curve $y = x^2$ and the line $y = x$ 1

(ii) Find any point(s) of intersection. 1

(iii) Find the volume of the solid formed when the area enclosed between the curve and the line is rotated about the x axis 3

QUESTION 4 (16 marks)



- (a) The diagram above shows the gradient function $y = f'(x)$ of a curve $y = f(x)$
- Describe the shape of the curve $y = f(x)$ where $x = A$ and $x = B$ 2
 - For what values of x is the curve $y = f(x)$ decreasing? 1
 - Sketch a possible curve $y = f(x)$, showing where $x = A, B$ and C . 2
- (b) For the function $y = 3 \cos 2x$:
- State the amplitude and the period. 2
 - Sketch $y = 3 \cos 2x$ in the domain $-\pi \leq x \leq \pi$ showing all important features, 3
- (c) A loan of \$50 000 is to be repaid in equal monthly instalments over 10 years.
- The interest rate is 12%p.a. reducible and each monthly repayment is \$P.
- If A_n is the amount owing at the end of the n th month, show that

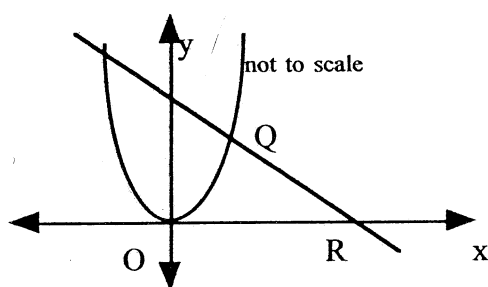
$$A_1 = 50000 \times 1.01 - P$$
 1
 - Find an expression for A_3 . 2
 - Find the amount of each monthly repayment. 3

SECTION C (start a new booklet)

QUESTION 5 (15 marks)

- (a)
- (i) On a number plane plot the points A(1,2), B(5,4) and C(7,8) 1
 - (ii) Show that the triangle ABC is isosceles. 2
 - (iii) The point D has coordinates (3,6). Show that ABCD is a rhombus. 2

(b)



- (i) Copy the diagram above of $y = x^2$ and $y = 6 - x$ into your booklet. 1
 - (ii) In the first quadrant, the line $y = 6 - x$ meets the x axis at R
and the curve intersects the line at Q. Show that the point Q
has coordinates (2, 4) 2
 - (iii) Find the coordinates of R. 1
 - (iv) Find the area enclosed by QR, the curve OQ and the x axis. 3
- (c) Use Simpson's Rule with 5 function values to find an approximation for
 $\int_0^4 e^x dx$ to 2 dec. pl. 3

QUESTION 6 (18 marks)

(a) Evaluate

(i) $\int (\cos x - \sin x) dx$ 1

(ii) $\int 3 \sec^2 2x dx$ 1

(iii) $\int_0^{\frac{\pi}{2}} 5 \sin \frac{x}{2} dx$ 3

(b) (i) Sketch the curve $y = e^x$ 1

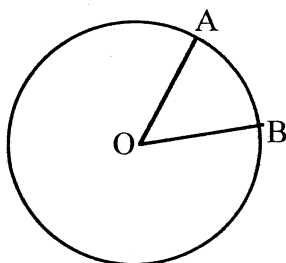
(ii) Find the coordinates of the point P where $x = a$ 1

(iii) Find the equation of the tangent at P. 2

(iv) Find the coordinates of the point T where the tangent cuts the x axis. 1

(iii) A perpendicular from P to the x axis meets the axis at N. Show that the distance TN is constant 2

(c)



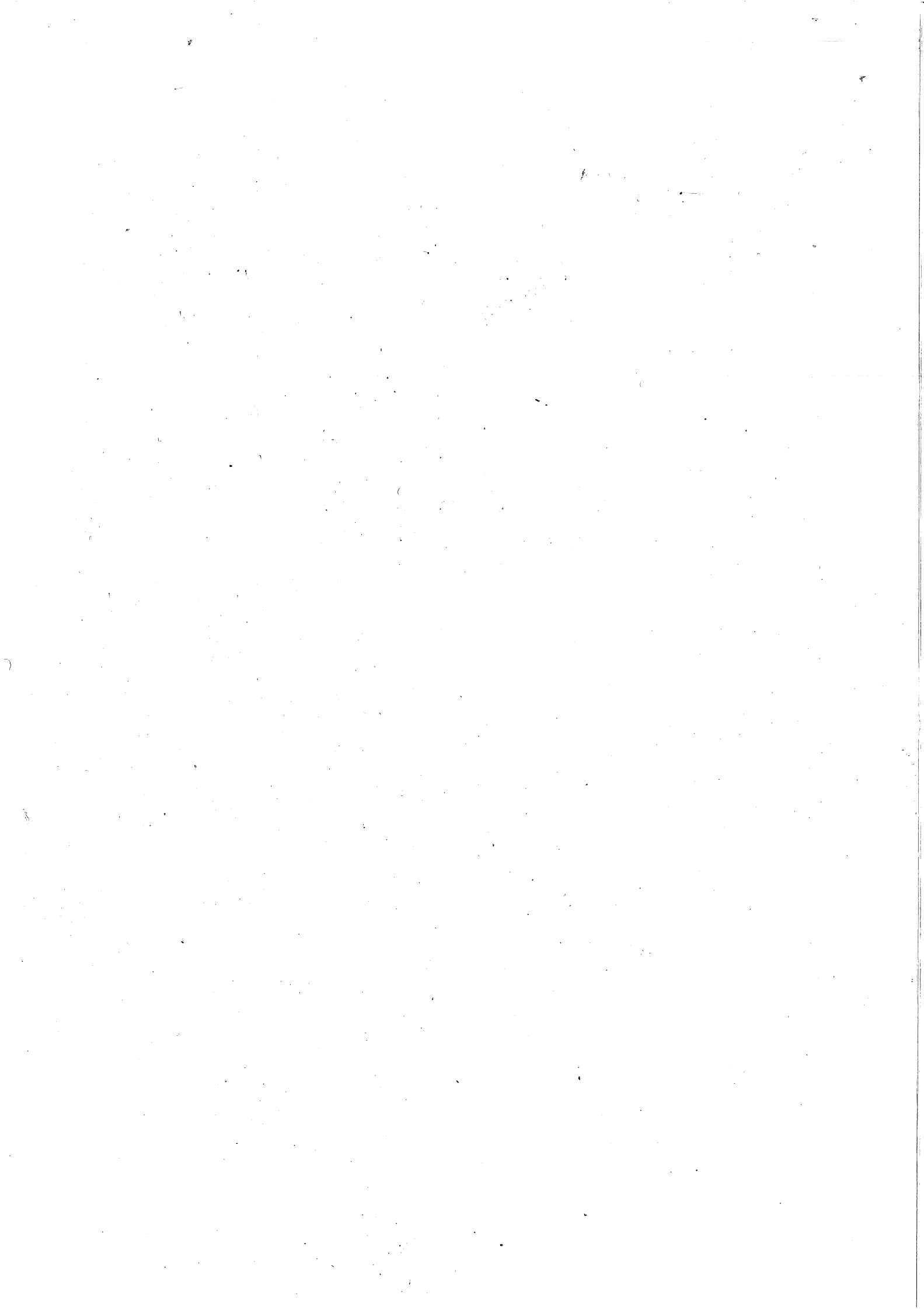
The circle above has radius 10 cm. The sector AOB has angle 45° at the centre.

(i) Find the length of the arc AB in terms of π . 1

(ii) The radii AO and OB are joined so that the sector AOB makes a cone.

Find the radius of the cone. 2

(iii) Find the volume of the cone correct to 2 dec. pl. 3



STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

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(1) (a) $-\sin 3x \times 3 = -3\sin 3x$ (1)

(b) $180^\circ = \pi$
 $1^\circ = \frac{\pi}{180}$

So $36 = 36 \times \frac{\pi}{180} = \frac{\pi}{5}$ (1)

(c) (i) $a = 2$ $d = 7$

$u_n = a + (n-1)d$

$u_{12} = 2 + 11 \times 7 = 79$ (1)

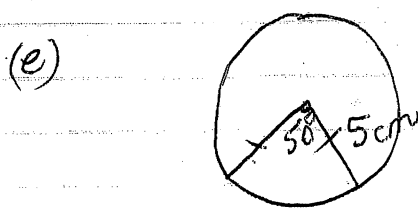
(ii) $S_n = \frac{n}{2}(2a + (n-1)d)$

$S_{15} = \frac{15}{2}(4 + 14 \times 7)$
 $= 765$ (2)

(d) domain $5 - 2x \geq 0$
 $-2x \geq -5$
 $x \leq \frac{-5}{-2}$

$x \leq 2\frac{1}{2}$ (1)

Range $y \geq 0$ (1)



$A = \frac{1}{2}r^2\theta$
 $= \frac{1}{2} \times 5^2 \times \frac{50\pi}{180}$

$= \frac{25 \times 50\pi}{2 \times 180}$

$= 3\frac{17}{36}\pi u^2 = \frac{125\pi}{36} u^2$ (2)

(f) (i) $\frac{1}{2}\cos x - \sin 2x \times 2 = \frac{1}{2}\cos x + 2\sin 2x$ (1)

(ii) $2\sin 2x \times \cos 2x \times 2 = 4\sin 2x \cos 2x$ (2)

(iii) $e^{3x-1} \times 3 - \frac{1}{2}e^{-2x-5} \times -2 = 3e^{3x-1} + e^{-2x-5}$ (2)

(f) (iv)

$\frac{\cos 2x \times 2 - 2x \times \sin 2x \times 2}{(\cos 2x)^2}$

$\frac{2\cos 2x + 4x\sin 2x}{(\cos 2x)^2}$ (2)

or as $2x(\cos 2x)^{-1}$

$\Rightarrow 2x \times -1(\cos 2x)^{-2} \times -2\sin 2x$
 $+ 2(\cos 2x)^{-1}$

$= \frac{+4x\sin 2x}{(\cos 2x)^2} + \frac{2}{(\cos 2x)^1}$

$= \frac{2\cos 2x + 4x\sin 2x}{(\cos 2x)^2}$

(2) (a)
$$\left. \begin{array}{l} n=1, \quad 3-4 = -1 \\ n=2, \quad 6-4 = 2 \\ n=3, \quad 9-4 = 5 \\ \vdots \\ n=20, \quad 60-4 = 56 \end{array} \right\} \begin{array}{l} a = -1 \\ d = 3 \\ L = 56 \\ n = 20 \end{array} \quad \begin{array}{l} S_n = \frac{n}{2}(a+L) \\ S_{20} = \frac{20}{2}(-1+56) \\ = 10 \times 55 = 550 \end{array} \quad (2)$$

(b) $0.\overset{0}{4}\overset{0}{3} = \frac{43}{99} \quad (2)$

(c) $T_1 = 3x \quad T_2 = 2x+4 \quad T_3 = 6x-2$

So $T_2 - T_1 = T_3 - T_2$
 $2x+4 - 3x = 6x-2 - (2x+4)$
 $-x+4 = 4x-6$
 $10 = 5x$
 $x = 2 \quad \{6, 8, 10\} \quad (2)$

(d) $\int \cos 3x - \sin 2x \, dx$

$= \frac{1}{3} \int 3 \cos 3x \, dx + \frac{1}{2} \int 2 \sin 2x \, dx = \frac{1}{3} \sin 3x + \frac{1}{2} \cos 2x + C \quad (1)$

(e) $\int 4(3x-4)^3 \, dx$
 $= 4 \int (3x-4)^3 \, dx = \frac{4(3x-4)^4}{4 \times 3} + C$

$\Rightarrow f(x) = \frac{(3x-4)^4}{3} + C$

$3 = \frac{(6-4)^4}{3} + C$

$C = 3 - \frac{16}{3} = -2\frac{1}{3}$

So $f(x) = \frac{(3x-4)^4}{3} - 2\frac{1}{3} \quad (2)$

(2) (f) $y = x^3 - 3x - 4$

$y' = 3x^2 - 3$

$y'' = 6x$

(i) Stat pts exist when $y' = 0$, $3x^2 - 3 = 0$

$3x^2 = 3$

$x^2 = 1$ $x = \pm 1$

When $x = 1$, $y = 1 - 3 - 4 = -6$ $(1, -6)$ ①

When $x = -1$, $y = -1 + 3 - 4 = -2$ $(-1, -2)$ ①

When $x = 1$, $y'' = 6 > 0$ min ①

$x = -1$, $y'' = -6 < 0$ max ①

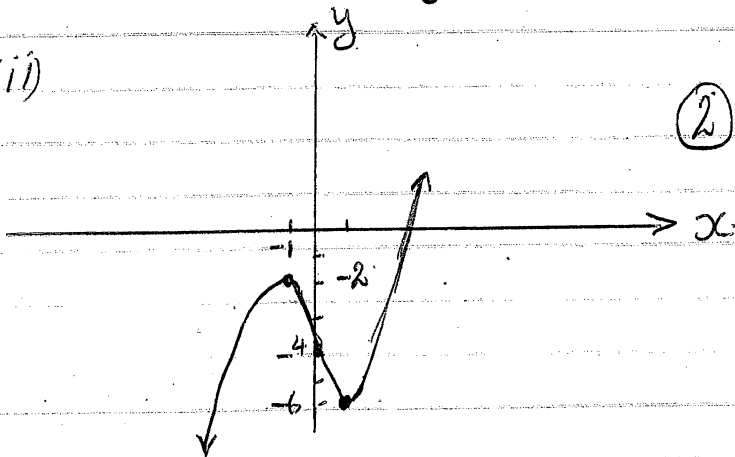
(ii) Inf occur when $y'' = 0$ and there is a sign change.

$y'' = 6x = 0$, $x = 0$

when $x = 0$, $y = 0 - 0 - 4$ $(0, -4)$

At $x = 0 - \epsilon$ $y'' < 0$ } sign change Yes, inflexion ①
 $x = 0 + \epsilon$ $y'' > 0$

(iii)



(iv) $x > 0$ ①

2 unit - Mathematics

Section B Solutions

Question 3

$$\begin{aligned} \text{a) i) } \int (2x - 3)(x - 1) dx \\ = \int (2x^2 - 2x - 3x + 3) dx \\ = \int (2x^2 - 5x + 3) dx \quad \textcircled{1} \end{aligned}$$

$$= \frac{2x^3}{3} - \frac{5x^2}{2} + 3x + C \quad \textcircled{1}$$

-1/2 for 1st inc eg
no +C.

$$\text{ii) } \int (4x - 1)^3 dx$$

$$= \frac{(4x - 1)^4}{4 \times 4} + C$$

$$= \frac{(4x - 1)^4}{16} + C$$

$$\text{iii) } \int_0^1 \frac{x^3 + 2x^2 - x}{x} dx$$

$$\int_0^1 \left(\frac{x^3}{x} + \frac{2x^2}{x} - \frac{x}{x} \right) dx$$

$$\int_0^1 (x^2 + 2x - 1) dx \quad \textcircled{1}$$

$$\left[\frac{x^3}{3} + \frac{2x^2}{2} - x \right]_0^1$$

$$= \left[\frac{1}{3} + 1 - 1 \right] - [0 + 0 - 0]$$

$$= \frac{1}{3} \quad \textcircled{1}$$

$$\text{iv) } \int_1^2 \frac{1+x}{\sqrt{x}} dx$$

$$= \int_1^2 \left(\frac{1}{\sqrt{x}} + \frac{x}{\sqrt{x}} \right) dx$$

$$= \int_1^2 (x^{-1/2} + x^{1/2}) dx$$

$$= \left[\frac{x^{1/2}}{1/2} + \frac{x^{3/2}}{3/2} \right]_1^2 \quad \textcircled{1}$$

$$= \left[2x^{1/2} + \frac{2x^{3/2}}{3} \right]_1^2$$

$$= \left[2 \times 2^{1/2} + \frac{2}{3} \times 2^{3/2} \right] -$$

$$\left[2 \times 1 + \frac{2}{3} \times 1 \right]$$

$$= \left[2\sqrt{2} + \frac{2\sqrt{8}}{3} \right] - \left[2 + \frac{2}{3} \right]$$

$$= \left[2\sqrt{2} + \frac{4\sqrt{2}}{3} \right] - \frac{8}{3}$$

$$= \frac{10\sqrt{2}}{3} - \frac{8}{3} \quad \textcircled{1}$$

$$= \frac{2}{3} (5\sqrt{2} - 4)$$

$$i) S_n = n^2 - 2n$$

$$i) S_1 = 1^2 - 2 \times 1 \\ = 1 - 2 \\ = -1 \quad (1)$$

$$ii) S_2 = 2^2 - 2 \times 2 \\ = 4 - 4 \\ = 0 \quad (1)$$

i) Formula for sum of an Arithmetic series

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$n = 2, a = -1, S_n = 0$$

$$0 = \frac{2}{2} [2 \times -1 + (2-1)d]$$

$$0 = 1 [-2 + d]$$

$$\therefore \boxed{d = 2}$$

$$S_3 = 3^2 - 2 \times 3 \\ = 3 \quad (2)$$

$$x + x + d + x + 2d = 3$$

$$3x + 3d = 3$$

$$-3 + 3d = 3$$

$$3d = 6$$

$$\therefore \boxed{d = 2}$$

Since we have proven that this series has

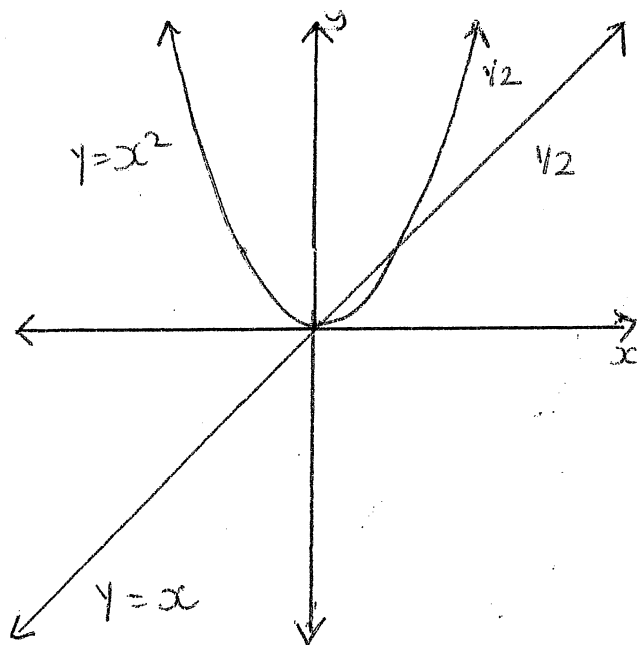
a common difference & not a common ratio

$S_n = n^2 - 2n$ is an Arithmetic series.

$$iv) T_n = a + (n-1)d$$

$$\therefore T_n = -1 + (n-1) \times 2 \quad (1)$$

c) i



$$ii) x = x^2$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$\therefore x = 0, x = 1 \quad (1)$$

$$y = 0$$

$$y = 1$$

or can read straight from an accurate graph.

iii)

$$I = \pi \int_0^1 x^2 dx - \pi \int_0^1 (x^2)^2 dx \quad (1)$$

$$= \pi \int_0^1 (x^2 - x^4) dx$$

$$= \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 \quad (1)$$

$$= \pi \left[\left(\frac{1}{3} - \frac{1}{5} \right) - (0) \right]$$

$$= \frac{2\pi}{15} \quad (1)$$

$1\frac{1}{2}$ for $\pi/30$. 1 for $\pi/6$

Question 4.

i) at $x=A$ as $f'(A)=0$

(a) has a zero gradient, (1)
 $\therefore$ a stationary point.

at $x=B$

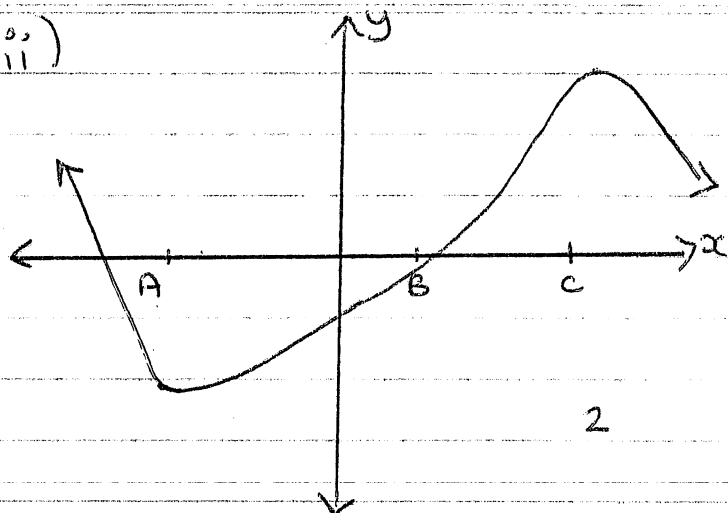
pt of inflexion

(1)

i) $f(x)$ is decreasing when
 $f'(x) < 0$

$\therefore x < A$ or $x > C$
 $\frac{1}{2}$ $\frac{1}{2}$

iii)



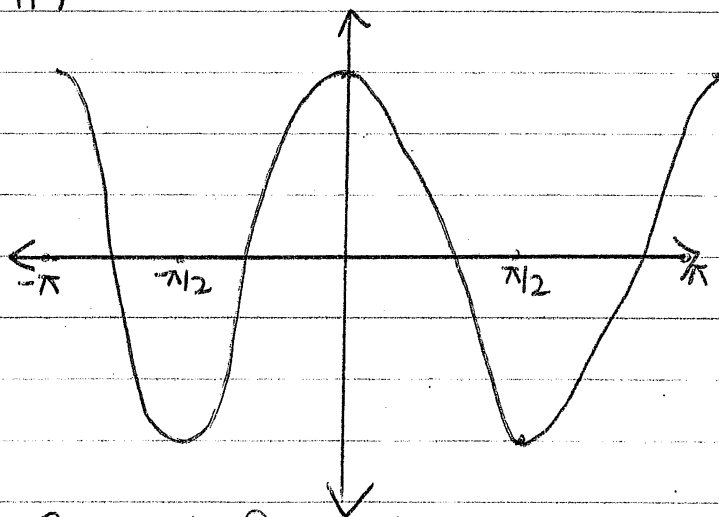
2

b) $y = 3 \cos 2x$

i) amplitude = 3 (1)

period $\frac{2\pi}{2} = \pi$ (1)

ii)



3 - 1 for 1st error
 $\frac{1}{2}$ for extra

c) \$50,000 loan
 monthly instalments 10 years $\therefore$ 120 payments.
 I R = 12% pa $\therefore$ 1% per month.
 repayment = \$P.

$$i) S_n = \frac{1.01(1.01^n - 1)}{1.01 - 1}$$

$$= 1.01.$$

Amount Owning
 = balance * interest
 - Payment. ①

$$A_1 = 50000 \times 1.01 - P.$$

~~$$i) A_3 = 50000 \left[\frac{1.01(1.01^3 - 1)}{1.01 - 1} \right] - P$$~~

$$A_2 = \frac{[50000 - P] \times 1.01 - P}{1.01 - 1}$$

$$= [50000 \times 1.01 - P] \times 1.01 - P$$

$$= 50000 \times (1.01)^2 - P(1.01 + 1)$$

$$\therefore A_3 = A_2 \times (1.01) - P$$

$$= [50000 \times 1.02^2 - P(1.01 + 1)](1.01) - P$$

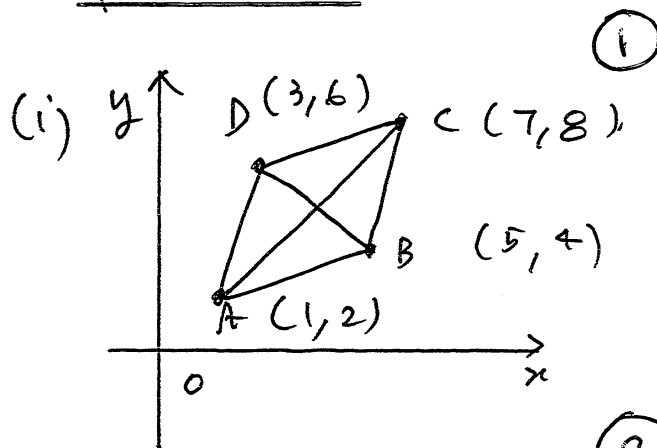
$$= 50000 \times 1.01^3 - P(1.01^2 + 1.01 + 1) \quad (2)$$

$$iii) A_{120} = \frac{50000 \times 1.01^{120} * 0.01}{1.01^{120} - 1}$$

$$= \$717.35 \text{ per month.} \quad (3)$$

15 (marks)

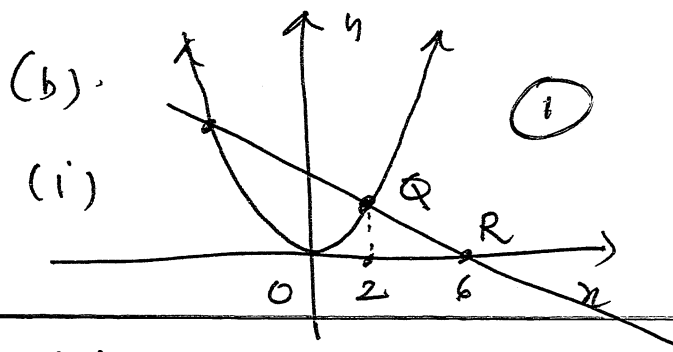
Question 5(a)



(ii) $AB = \sqrt{(-4)^2 + 4} = 2\sqrt{5}$
 $BC = \sqrt{(-2)^2 + (-4)^2} = 2\sqrt{5}$

$\therefore |AB| = |BC| = 2\sqrt{5}$
 i.e. $\triangle ABC$ is isosceles

(iii) $|CD| = |AD| = 2\sqrt{5}$
 i.e. $|AB| = |BC| = |CD| = |DA|$
 i.e. ABCD is rhombus



(ii) $x^2 = 6 - x$
 $\therefore x^2 + x - 6 = 0$
 $\therefore (x+3)(x-2) = 0$
 $x = -3 \quad x = 2$
 $y = 9 \quad y = 4$
 $\therefore Q(2, 4)$

(iii) $y = 6 - x, \quad 0 = 6 - x$
 $\Rightarrow R(6, 0)$

(iv) $A = \int_0^2 x^2 dx + \int_2^6 (6-x) dx$
 $= \left[\frac{x^3}{3} \right]_0^2 + \left[6x - \frac{x^2}{2} \right]_2^6$
 $= \frac{8}{3} + [18 - 10]$
 $= 10\frac{2}{3} \quad (3\frac{2}{3})$

(c) $\int_0^4 e^x dx$
 $= \frac{1}{3} [(y_1 + y_5) + 2y_2 + 4(y_3 + y_4)]$

x	y/x	$f(x) = e^x$
0	y_1	1
1	y_2	2.7183
2	y_3	7.3891
3	y_4	20.0855
4	y_5	54.5982

$= \frac{1}{3} [55.5982 + 14.7782 + 91.2152]$
 $= 53.8639$

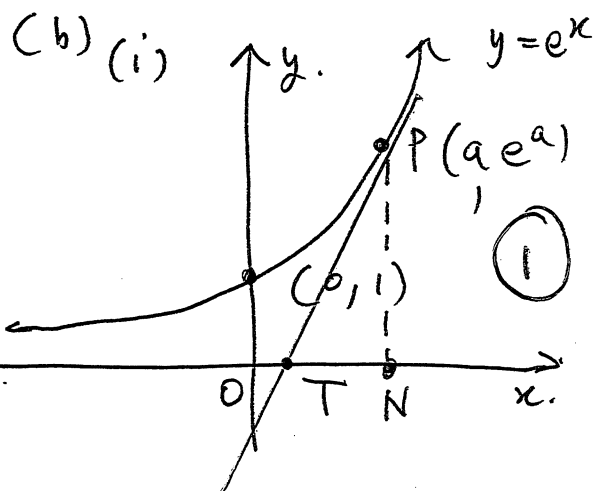
18 marks

Question 6 (a)

(i) $\int (\cos x - \sin x) dx$
 $= \sin x + \cos x + c$

(ii) $3 \int \sec^2 2x dx$
 $= \frac{3}{2} \tan 2x + c.$

(iii) $\int_0^{\pi/2} \frac{1}{5} \sin \frac{x}{2} dx$
 $= \left[-10 \cos \frac{x}{2} \right]_0^{\pi/2}$
 $= 10 \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)$
 $= \frac{5(2-\sqrt{2})}{2} \approx (2.93)$



(ii) $(a, e^a) = P.$ (1)

(iii) $\frac{dy}{dx} \Big|_{x=a} = e^a.$

$\therefore y - e^a = e^a(x - a)$

$y = (e^a)x - e^a(a-1)$ (2)

(iv) $y = 0.$

$e^a(a-1) = e^a x$

$\therefore x = (a-1)$

$T(a-1, 0).$ (1)

$N(a, 0).$

$\therefore |TN| = 1$ (which is constant).

(2)

(c)

(i) arc AB $= 10 \times \frac{\pi}{4}$
 $= 5\pi/2$ cm. (1)

(ii) $C = 2\pi r$
 $C = 10 \times (2\pi - \frac{\pi}{4})$
 i.e. $\frac{35\pi}{2} = 2\pi r$ (2)
 $r = \frac{35}{4} = 8.75$ cm.

(iii) $r^2 + 8.75^2 = 100$
 $\therefore r = 4.84$
 $V = \frac{1}{3} \pi \times (8.75)^2 \times 4.84$
 $= 388.15$ cm³. (3)